

Volume II · Chapter 9 — Medical Domain Adaptation · Labs

Source: split from `med-tech-curriculum-V1.md` (V1). From this split onward this chapter is maintained **independently** here. See `Volume-II-Chapter-9-context.md` for the AI interaction log.

Prerequisites: Ch 7, Ch 8.

Learning outcomes — the student can: distinguish continued pretraining vs. fine-tuning vs. instruction tuning; explain how Meditron/Clinical-Camel/MedGemma were adapted; choose an adaptation strategy; reason about data requirements and licensing.

Labs (hands-on) — 6 h:

#	Lab	h
9a	Compare a general vs. domain-adapted model on medical prompts	2
9b	Prepare a small biomedical instruction dataset (format, clean)	2
9c	Design an adaptation plan for a chosen clinical use case	2

Datasets/tools: HF models (Meditron/MedGemma/PubMedBERT), datasets. **Assessment:** adaptation-plan deliverable (**50%**); comparison lab (**30%**); quiz (**20%**). **Key decisions:** pretrain vs. fine-tune vs. RAG; data licensing; base-model choice. **References:** plan §13 → *Medical LLMs; Fine-tuning, alignment & benchmarks*. **Hours:** Theory **10** + Lab **6** = **16**.